O Princípio da Homogeneidade e Análise Dimensional

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Homogeneidade dimensional

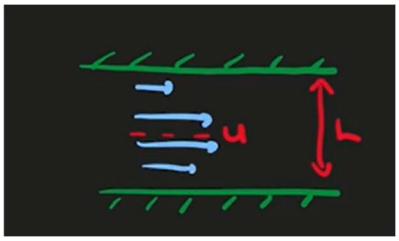
Dimensions Associated with Common Physical Quantities

	FLT System	*MLT* System		*FLT* System	*MLT* System
Acceleration	LT^{-2}	LT^{-2}	Momentum	FT	MLT^{-1}
Angle	$F^0L^0T^0$	$M^0L^0T^0$	Power	FLT^{-1}	ML^2T^{-3}
Angular acceleration	T^{-2}	T^{-2}	Pressure	FL^{-2}	$ML^{-1}T^{-2}$
Angular velocity	T^{-1}	T^{-1}	Specific heat	$L^2T^{-2}\Theta^{-1}$	$L^2T^{-2}\Theta^{-1}$
Area	L^2	L^2	Specific weight	FL^{-3}	$ML^{-2}T^{-2}$
Density	$FL^{-4}T^2$	ML^{-3}	Strain	$F^0L^0T^0$	$M^0L^0T^0$
Energy	FL	ML^2T^{-2}	Stress	FL^{-2}	$ML^{-1}T^{-2}$
Force	F	MLT^{-2}	Surface tension	FL^{-1}	MT^{-2}
Frequency	T^{-1}	T^{-1}	Temperature	Θ	Θ
Heat	FL	ML^2T^{-2}	Time	T	T
Length	L	L	Torque	FL	ML^2T^{-2}
Mass	$FL^{-1}T^2$	M	Velocity	LT^{-1}	LT^{-1}
Modulus of elasticity	FL^{-2}	$ML^{-1}T^{-2}$	Viscosity (dynamic)	FTL^{-2}	$ML^{-1}T^{-1}$
Moment of a force	FL	ML^2T^{-2}	Viscosity (kinematic)	L^2T^{-1}	L^2T^{-1}
Moment of inertia (area)	L^4	L^4	Volume	L^3	L^3
Moment of inertia (mass)	FLT^2	ML^2	Work	FL	ML^2T^{-2}

Variáveis, parâmetros ou números adimensionais

$\rho {V_0}^2 L^2$ V_0 L

$V_0 = u$ $L = b$



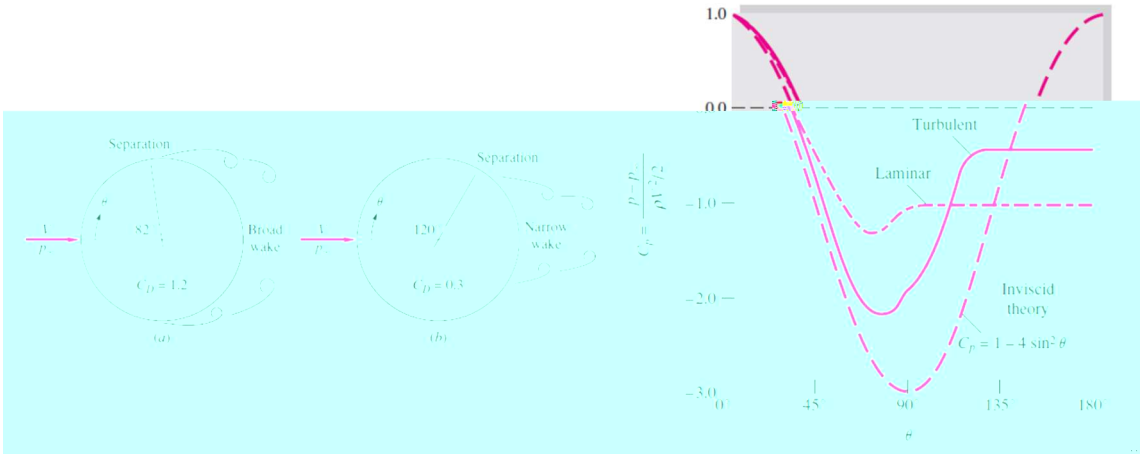
K_e

$K_e \cong mu^2$

$K_e = F.L$

$$F_{inércia} = \frac{mu^2}{L} = \frac{\rho u^2 L^3}{L} = \rho u^2 L^2$$

O número de Euler



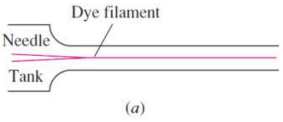
O número de Euler

$$\Delta p = \frac{F}{A} = \frac{F}{L^2} \quad \rightarrow F_{pressão} = \Delta p L^2$$

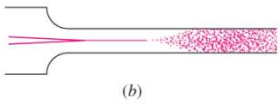
$$\frac{F_{pressão}}{F_{inércia}} = \frac{\Delta p L^2}{\rho {V_0}^2 L^2} = \frac{\Delta p}{\rho {V_0}^2}$$

$$Eu$$

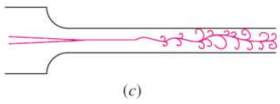
$$Eu = \frac{\Delta p}{\rho {V_0}^2}$$



(a) Regime laminar.



(b) Regime turbulento.



(c) Regime turbulento, foto instantânea. $(Re_D)_{cr} \cong 2,3 \times 10^3$ para dutos comerciais de diâmetro D.

$$\tau = \frac{F_{viscosa}}{A} \rightarrow F_{viscosa} = \tau A$$

$$\tau = \mu \frac{\Delta u}{\Delta y}$$

$$F_{viscosa} = \left(\mu \frac{\Delta u}{\Delta y}\right) A$$

$$\Delta u \rightarrow u \quad \Delta y \rightarrow L \quad A \rightarrow L^2$$

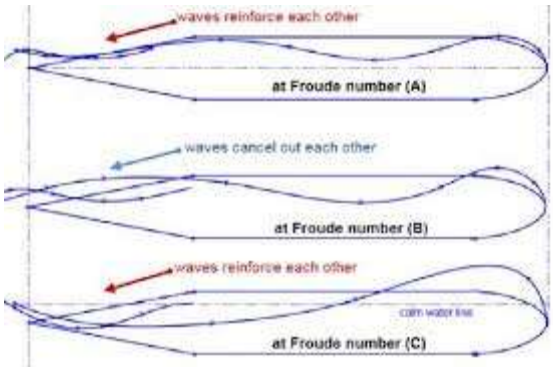
$$F_{viscosa} = \mu \frac{uL^2}{L}$$

$$\frac{F_{inércia}}{F_{viscosa}} = \frac{\rho u^2 L^2}{\mu \frac{V_0 L^2}{L}} = \frac{\rho u L}{\mu}$$

$$Re$$

$$Re = \frac{\rho u L}{\mu}$$

O número de Froude



$$F_{gravidade} = mg = \rho.Vol.g$$

$$F_{gravidade} = \rho L^3 g$$

$$\frac{F_{inércia}}{F_{gravidade}} = \frac{\rho {V_0}^2 L^2}{\rho L^3 g} = \frac{{V_0}^2}{gL}$$

Fr

$$Fr = \sqrt{\frac{{V_0}^2}{gL}} = \frac{V_0}{\sqrt{gL}}$$

O número de Weber

$$F_{superf} = \sigma L$$

σ

$$\frac{F_{inércia}}{F_{superf}} = \frac{\rho u^2 L^2}{\sigma L} = \frac{\rho u^2 L}{\sigma}$$

$$Wn = \frac{\rho u^2 L}{\sigma}$$

Análise dimensional

-
-
-

O teorema de Buckingham (1914)

u A $Q = uA$

p, u, ρ, L

$p: [ML^{-1}T^{-2}], u: [LT^{-1}], \rho: [ML^{-3}], L: [L]$

$\Delta p, u, \rho, L$

u, ρ, L

$\pi_1 = \Delta p u^a \rho^b L^c$

a, b e c

$[M^0L^0T^0] = [ML^{-1}T^{-2}]\,[LT^{-1}]^a[ML^{-3}]^b[L]^c$

$[M^0L^0T^0] = [M]^{1+b}[L]^{-1+a-3b+c}[T]^{-2-a}$

$M: 0 = 1 + b \quad \rightarrow \quad b = -1$

$L: 0 = -1 + a - 3b + c$

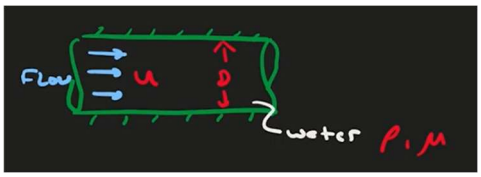
$T: 0 = -2 - a \quad \rightarrow \quad a = -2$

$c = 0$

$\pi_1 = \Delta p u^{-2} \rho^{-1} L^0$

π_1 Eu

$\pi_1 = Eu = \dfrac{\Delta p}{\rho {V_0}^2}$



u, D, μ, ρ

$u: [LT^{-1}], D: [L], \mu: [ML^{-1}T^{-1}] \rho: [ML^{-3}]$

u, μ, ρ

$\pi_1 = D\rho^a \mu^b u^c$

a, b e c

$[M^0L^0T^0] = [L]\,[ML^{-3}]^a [ML^{-1}T^{-1}]^b [LT^{-1}]^c$

$[M^0L^0T^0] = [M]^{a+b}[L]^{1-3a-b+c}[T]^{-b-c}$

$M: 0 = a + b \quad \rightarrow \quad a = -b$

$L: 0 = 1 - 3a - b + c$

$T: 0 = -b - c \quad \rightarrow \quad c = -b$

$c = 0$

$a = -b \qquad\qquad c = -b$

$L: 0 = 1 - 3a - b + c \quad \rightarrow 1 + 3b - b - b = 0 \rightarrow \quad b = \quad -1$.

Então $a = 1$ e $c = 1$ e

$\pi_1 = D\rho^1 \mu^{-1} u^1$

π_1 Re

$$\pi_1 = Re = \frac{\rho u D}{\mu}$$

Os coeficientes de Resistência de Forma, Resistência Friccional e Resistência de Ondas na hidrodinâmica do navio

Determinação do π_1

$$\pi_1 = R_{forma} \rho^a L^b U^c$$

$$[M^0 L^0 T^0] = [M^1 L^1 T^{-2}][M^1 L^{-3}]^a [L^1]^b [L^1 T^{-1}]^c$$

$$0 = M^{1+a} L^{1-3a+b+c} T^{-2-c}$$

$$\pi_1 = R_{forma}\rho^{-1}L^{-2}U^{-2}$$

$$\pi_1 = \frac{R_{forma}}{\rho L^2 U^2}$$

Determinação do π_2

$$\pi_2 = \mu\rho^a L^b U^c$$

$$[M^0L^0T^0] = [M^1L^{-1}T^{-1}][M^1L^{-3}]^a[L^1]^b[L^1T^{-1}]^c$$

$$0 = M^{1+a}L^{-1-3a+b+c}T^{-1-c}$$

$$\pi_2 = \mu\rho^{-1}L^{-1}U^{-1}$$

$$\pi_2 = \frac{\mu}{\rho \mathrm{L} U}$$

$$\pi_2 = Rn^{-1}$$

π_1 π_2

Determinação do π_3

$$\pi_3 = g\rho^a L^b U^c$$

$$[M^0L^0T^0] = [L^1T^{-2}][M^1L^{-3}]^a[L^1]^b[L^1T^{-1}]^c$$

$$0 = M^aL^{1-3a+b+c}T^{-1-c}$$

$$\pi_3 = g\rho^0L^1U^{-2}$$

$$\pi_3 = \frac{gL}{U^2}$$

$$\pi_3 = Fn^{-2}$$

Determinação do π_4

$$\pi_4 = \sigma\rho^aL^bU^c$$

$$[M^0L^0T^0] = [MT^{-2}][M^1L^{-3}]^a[L^1]^b[L^1T^{-1}]^c$$

$$0 = M^{1+a}L^{-3a+b+c}T^{-2-c}$$

$$\pi_4 = \sigma\rho^{-1}L^{-1}U^{-2}$$

$$\pi_4 = \frac{\sigma}{\rho LU^2}$$

$$\pi_4 = Wn^{-1}$$

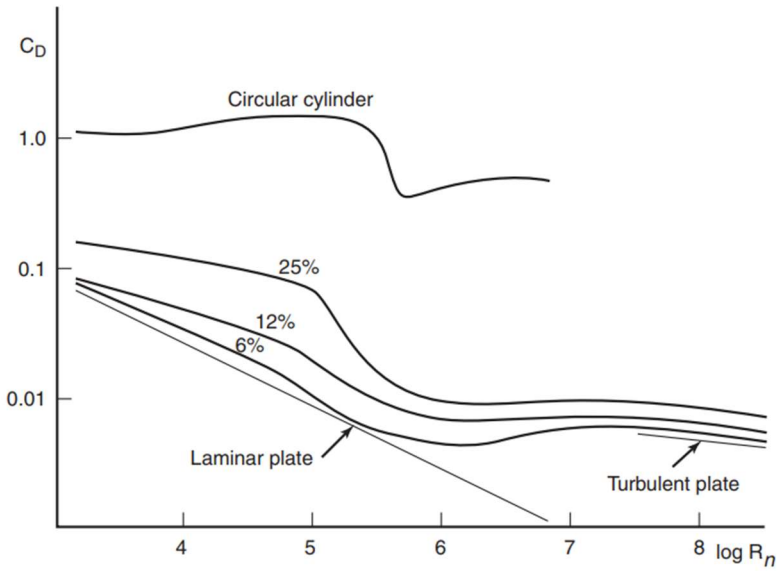
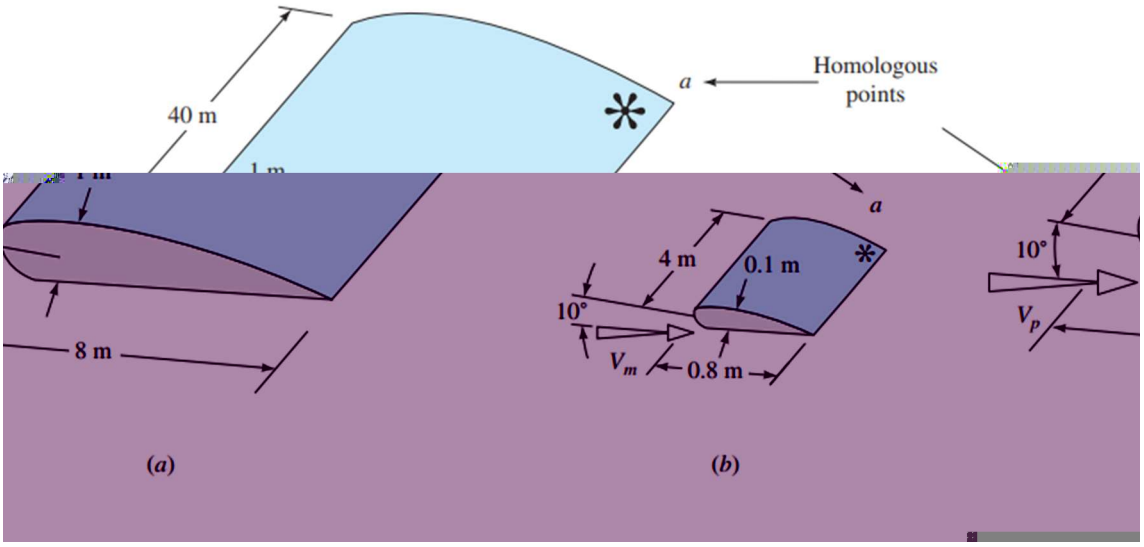
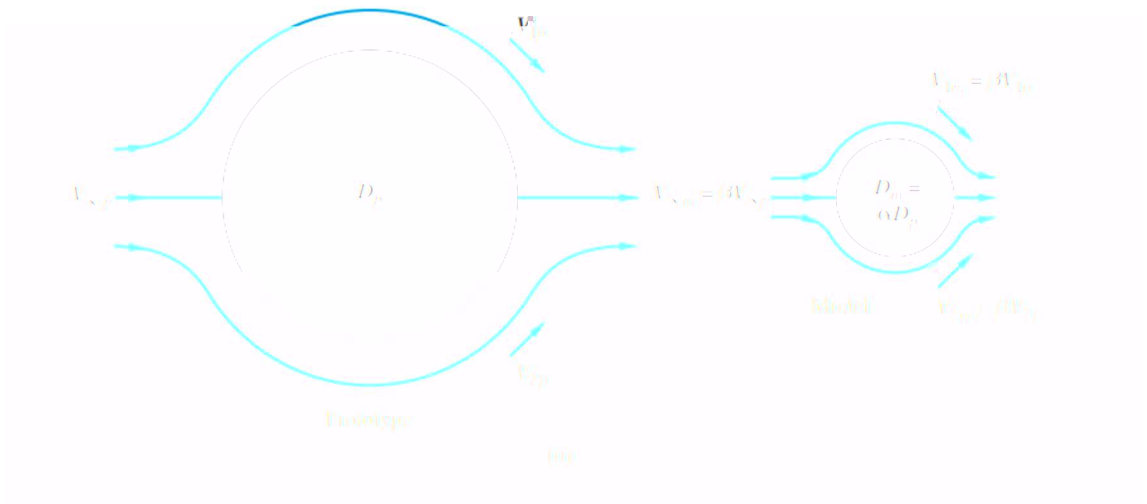


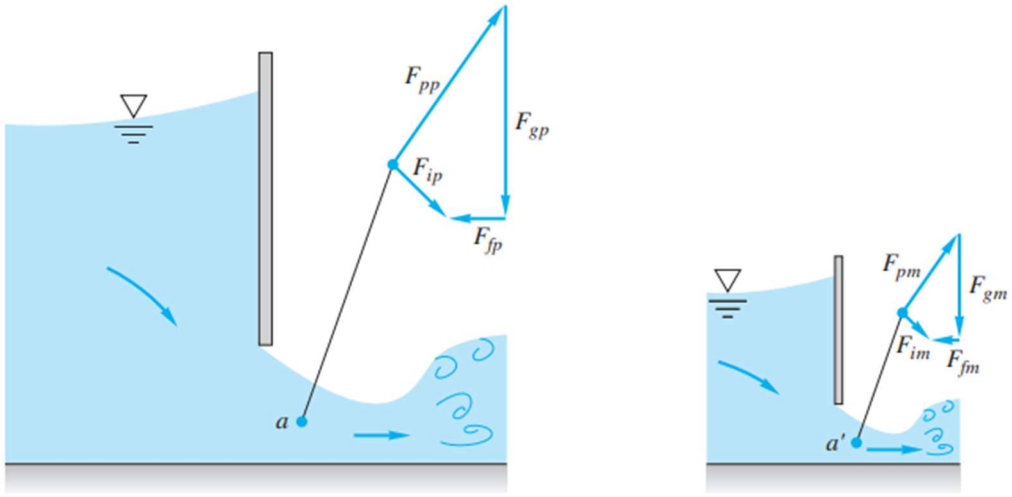
Figure 7.18 Reynolds number dependence of section drag.

O conceito da similaridade

1. The model nose radius must be one-tenth as large.
2. The model surface roughness must be one-tenth as large.




Prototype
Model



(a) (b)

O dilema da similaridade em testes com modelos

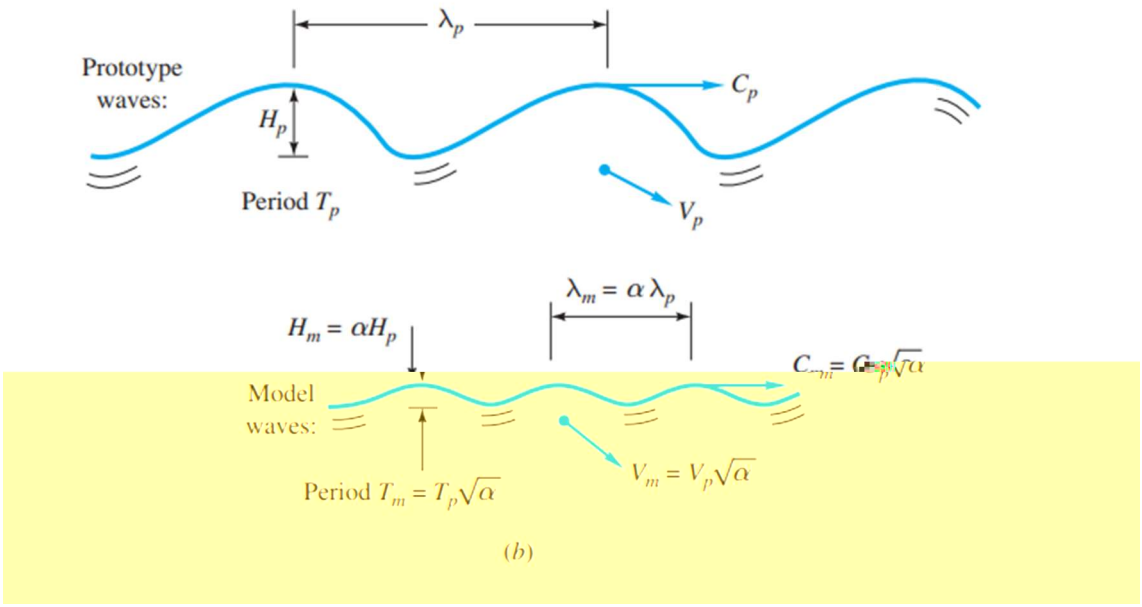
$$Fr_m = \frac{{V_m}^2}{gL_m} = \frac{{V_p}^2}{gL_p} = Fr_p$$

α

$$L_m = \quad L_p\alpha$$

$$\frac{{V_m}^2}{{V_p}^2} = \frac{L_m}{L_p} \quad \rightarrow \quad \frac{V_m}{V_p} = \sqrt{\frac{L_m}{L_p}} = \sqrt{\alpha}$$

$$\frac{T_m}{T_p} = \frac{L_m/V_m}{L_p/V_p} = \frac{\alpha}{\sqrt{\alpha}} = \sqrt{\alpha}$$



$$\frac{V_m L_M}{\nu_m} = \frac{V_p L_p}{\nu_p}$$

$$\frac{\nu_m}{\nu_p} = \frac{L_M V_m}{L_p V_p} = \alpha\sqrt{\alpha} = \alpha^{3/2}$$

$$\alpha = 0.1 \quad \alpha^{3/2} = 0.032$$

Fluid	μ, kg/(m · s)[†]	Ratio $\mu/\mu(H_2)$	ρ, kg/m^3	ν m^2/s[†]	Ratio ν/ν(Hg)
Hydrogen	9.0 E–6	1.0	0.084	1.05 E–4	910
Air	1.8 E–5	2.1	1.20	1.50 E–5	130
Gasoline	2.9 E–4	33	680	4.22 E–7	3.7
Water	1.0 E–3	114	998	1.01 E–6	8.7
Ethyl alcohol	1.2 E–3	135	789	1.52 E–6	13
Mercury	1.5 E–3	170	13,550	1.16 E–7	1.0
SAE 30 oil	0.29	33,000	891	3.25 E–4	2,850
Glycerin	1.5	170,000	1,260	1.18 E–3	10,300

[†]1 kg/(m · s) = 0.0209 slug/(ft · s); 1 m^2/s = 10.76 ft^2/s.